=== UNIT 5: OPERATIONAL AMPLIFIERS (OP-AMPS) & ACTIVE FILTERS ===

>> 5.1 Operational Amplifier Architecture and Ideal Characteristics

An Operational Amplifier (Op-Amp) is a high-gain DC-coupled differential voltage amplifier designed to perform mathematical operations (addition, subtraction, integration, differentiation) using negative feedback.
* Op-Amp Terminals: Non-inverting input (+), Inverting input (-), Output (V_out), and DC power rails (+V_CC, -V_EE).
* Ideal Op-Amp Parameters (Benchmark Comparison):
1. Differential Voltage Gain A_vd = infinity (Practically ~ 10^5 to 10^6).
2. Input Impedance R_in = infinity (Practically ~ 10^6 to 10^12 Omega, zero input current: I+ = I- = 0).
3. Output Impedance R_out = 0 (Practically ~ 50 - 100 Omega).
4. Bandwidth BW = infinity (Can amplify signals from DC up to gigahertz).
5. Common Mode Rejection Ratio (CMRR) = infinity (Rejects all common-mode electrical noise).
6. Slew Rate SR = infinity (Instantaneous output voltage change: dV_out/dt).
7. Input Offset Voltage V_os = 0 (V_out = 0 when V+ = V-).
* Concept of Virtual Ground: When negative feedback is applied, infinite open-loop gain forces the differential input voltage to zero: V_d = V+ - V- = V_out / A_vd = V_out / infinity = 0. Therefore, V- = V+. If the non-inverting terminal is connected to physical ground (V+ = 0V), the inverting terminal is held at virtual ground potential (V- = 0V) without a direct physical ground connection.

>> 5.2 Linear Op-Amp Configurations

* Inverting Amplifier: Input applied to inverting terminal through resistor R1, feedback resistor Rf connects output to inverting terminal; V+ = 0. By KCL at virtual ground: I_in = I_f => (V_in - 0)/R1 = (0 - V_out)/Rf => Closed-Loop Gain A_v = V_out / V_in = - Rf / R1. Phase shift is 180 degrees.
* Non-Inverting Amplifier: Input applied directly to non-inverting terminal (V+ = V_in). Voltage divider formed by R1 and Rf feeds back to inverting terminal: V- = V_out * [R1 / (R1 + Rf)]. Setting V- = V+: V_out = V_in * [1 + Rf / R1]. Phase shift is 0 degrees.
* Voltage Follower (Buffer): Non-inverting amplifier with Rf = 0 and R1 = infinity: A_v = +1. Features ultra-high input impedance and very low output impedance, isolating delicate biosensors from low-impedance recording loads.
* Summing Amplifier (Adder): Multiple inputs V1, V2, V3 connected through input resistors to inverting node: V_out = - [(Rf/R1)*V1 + (Rf/R2)*V2 + (Rf/R3)*V3].
* Difference Amplifier (Subtractor): V_out = (Rf / R1) * (V2 - V1), provided resistor matching R3/R2 = Rf/R1 is satisfied.

>> 5.3 Active Wave-Shaping: Integrators and Differentiators

* Op-Amp Integrator: Feedback resistor is replaced by a capacitor C. By KCL at virtual ground: $I_R = I_C \Rightarrow V_{in} / R = - C * (dV_{out} / dt)$.
$V_{out}(t) = - (1 / (R * C)) * \int_0^t V_{in}(\tau) d\tau + V_{initial}$.
Converts square waves to triangular waves; acts as an active low-pass filter.
* Op-Amp Differentiator: Input element is capacitor C, feedback element is resistor R: $I_C = I_R \Rightarrow C * (dV_{in} / dt) = - V_{out} / R$.
$V_{out}(t) = - R * C * (dV_{in} / dt)$.
Converts triangular waves to square spikes; prone to high-frequency noise amplification.

[SOLVED PROBLEM 1: Inverting and Non-Inverting Amplifier Design]

Problem: Design an inverting amplifier with a voltage gain of $A_v = -12$ and an input impedance of 10 kΩ. If the op-amp is powered by +/- 15V rails and an input of V_{in} = 1.0 V is applied, calculate: (1) Values of R1 and Rf, (2) Output voltage V_{out}, (3) Output voltage if V_{in} = 1.5 V.
Solution Step 1: Component Values:
Input impedance of an inverting amplifier equals R1. Therefore: R1 = 10 kΩ.
Gain magnitude $|A_v| = Rf / R1 = 12 \Rightarrow Rf = 12 * R1 = 12 * 10$ kΩ = 120 kΩ.
Solution Step 2: Output Voltage for V_{in} = 1.0 V:
$V_{out} = A_v * V_{in} = (-12) * (1.0 V) = -12.0$ Volts.
Since -12V is within the linear range (-15V to +15V, accounting for saturation limit ~ +/-13.5V), output is undistorted.
Solution Step 3: Output Voltage for V_{in} = 1.5 V:
Theoretical $V_{out} = (-12) * (1.5 V) = -18.0$ Volts.
Saturation Check: The output cannot exceed the negative supply rail (-15V). The op-amp enters negative saturation.
Actual Output: V_{out} ~ -13.5 V to -14.0 V (Clipped at negative saturation).

[SOLVED PROBLEM 2: Precision Summing Amplifier Calculation]

Problem: An inverting summing amplifier has Rf = 100 kΩ, with three input resistors R1 = 20 kΩ, R2 = 50 kΩ, and R3 = 10 kΩ connected to voltages V1 = +0.5 V, V2 = -1.0 V, and V3 = +0.2 V. Determine the output voltage V_{out}.
Solution Step 1: Formula for Summing Amplifier:
$V_{out} = - [(Rf / R1) * V1 + (Rf / R2) * V2 + (Rf / R3) * V3]$.
Solution Step 2: Compute individual gain weighting factors:
Gain 1: Rf / R1 = 100 kΩ / 20 kΩ = 5.0.
Gain 2: Rf / R2 = 100 kΩ / 50 kΩ = 2.0.
Gain 3: Rf / R3 = 100 kΩ / 10 kΩ = 10.0.

Solution Step 3: Substitute inputs:
V_out = - [5.0 * (0.5 V) + 2.0 * (-1.0 V) + 10.0 * (0.2 V)]
V_out = - [2.5 - 2.0 + 2.0] = - [2.5] = -2.5 Volts.
Final Answer: V_out = -2.5 Volts.

[SOLVED PROBLEM 3: Practical Active Integrator Waveform Response]

Problem: An ideal Op-Amp integrator has R = 100 kOmega and C = 0.1 microF. A constant DC step voltage of V_in = +2.0 V is applied at t = 0 with initial capacitor voltage V_c(0) = 0. Determine: (1) Integrator time constant tau, (2) Rate of change of output voltage, (3) Time required for the output to saturate at -13V.
Solution Step 1: Compute Time Constant tau:
tau = R * C = (100 * 10^3 Omega) * (0.1 * 10^-6 F) = 0.01 seconds = 10 milliseconds.
Solution Step 2: Rate of Change of Output Voltage dV_out/dt:
dV_out / dt = - V_in / (R * C) = - 2.0 V / 0.01 s = -200 Volts/second.
Solution Step 3: Output Voltage Function:
V_out(t) = - 200 * t Volts (a downward linear ramp).
Solution Step 4: Time to Reach Saturation at V_out = -13 V:
-13 = - 200 * t => t_sat = 13 / 200 = 0.065 seconds = 65 milliseconds.
Conclusion: Output decreases linearly at 200 V/s until clipping at negative saturation at t = 65 ms.

[SOLVED PROBLEM 4: Instrumentation Difference Amplifier Analysis]

Problem: In a bio-potential differential amplifier circuit, R1 = R2 = 10 kOmega, and feedback resistors Rf = R4 = 100 kOmega. Common mode voltage V_cm = 5.0 V with a biological differential ECG signal V_d = 20 mV (V1 = 5.010 V, V2 = 4.990 V). Verify that the common mode voltage is rejected and calculate V_out.
Solution Step 1: Verify Balanced Resistor Condition for Ideal CMRR:
Condition: R4 / R2 = Rf / R1 => 100k / 10k = 100k / 10k = 10.0 (Condition strictly satisfied).
Solution Step 2: Calculate Output Voltage:
V_out = (Rf / R1) * (V1 - V2) = 10.0 * (5.010 V - 4.990 V) = 10.0 * (0.020 V) = 0.200 Volts = 200 mV.
Common Mode Rejection Check: The 5V DC common-mode bias is completely eliminated; only the 20 mV ECG biopotential is amplified 10-fold to 200 mV.

>> 5.4 University Review & Examination Questions

* Q1: State the characteristics of an ideal Op-Amp and explain the concept of Virtual Ground in detail.

Answer: Ideal characteristics: (1) Infinite open-loop differential gain A_vd = inf, (2) Infinite input impedance R_in = inf (zero input current), (3) Zero output impedance R_out = 0, (4) Infinite bandwidth, (5) Infinite CMRR, (6) Infinite Slew Rate, (7) Zero offset voltage. Concept of Virtual Ground: When negative feedback is applied, the high open-loop gain ensures the output voltage remains within supply boundaries: V_out = A_vd * (V+ - V-). Since A_vd -> inf, differential input voltage must be virtually zero: V+ - V- = V_out / A_vd = 0 => V- = V+. If the non-inverting terminal (+) is tied to ground (V+ = 0), the inverting terminal (-) is maintained at 0V potential. It acts as a ground for all circuit calculations, yet no physical path exists to ground (no current flows into the pin due to infinite input impedance).

* Q2: Explain the operation and frequency response limitations of an Op-Amp differentiator, and describe how practical differentiators overcome instability.

Answer: An ideal differentiator has input capacitor C and feedback resistor R, yielding V_out = - R*C*(dV_in/dt). Its transfer function is H(s) = - s*R*C, which has a magnitude |H(j*omega)| = omega*R*C that increases linearly with frequency at +20 dB/decade without limit. This causes two severe practical problems: (1) High-frequency electrical noise is drastically magnified, obscuring desired signals, (2) Phase lag causes closed-loop instability and high-frequency oscillations. A practical differentiator adds a small series resistor R_s at the input and a small parallel capacitor C_f across feedback resistor R. At high frequencies, R_s and C_f attenuate signals, converting the circuit into an integrator/low-pass filter above a critical corner frequency f_c = 1 / (2*pi*R_s*C), stabilizing the loop.

>> 5.5 Key Takeaways and Revision Points

* Ideal Op-Amp parameters: A_v = inf, R_in = inf, R_out = 0, CMRR = inf.
* Inverting amplifier gain: A_v = - Rf / R1; Non-inverting amplifier gain: A_v = 1 + Rf / R1.
* Virtual ground holds inverting terminal at 0V when non-inverting terminal is grounded.
* Op-Amp Integrator uses feedback capacitor to integrate input waveforms over time.